MSG No. 64.478 C_cmca [Type IV, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[1,3,18,20]`
2	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[2,4,17,19]`
3	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[5,7,22,24]`
4	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[6,8,21,23]`
5	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,11,26,28]`
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[10,12,25,27]`
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,15,30,32]`
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[14,16,29,31]`

Table 2: Wyckoff site: `8b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, 0\right]$	`[1,2,19,20]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[3,4,17,18]`
3	$\left[\frac{1}{4}, 0, 0\right]$	`[5,6,23,24]`
4	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[7,8,21,22]`
5	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[9,10,27,28]`
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,25,26]`
7	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[13,14,31,32]`
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,16,29,30]`

Table 3: Wyckoff site: `8c`, site symmetry: `.2/m'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,3,29,31]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,4,30,32]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[5,7,25,27]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,8,26,28]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,11,21,23]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[10,12,22,24]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[13,15,17,19]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[14,16,18,20]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,13,19,31]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,14,20,32]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,15,17,29]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,16,18,30]`
5	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[5,9,23,27]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[6,10,24,28]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[7,11,21,25]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[8,12,22,26]

Table 5: Wyckoff site: 8e, site symmetry: 2'/m..

No.	position	mapping
1	$\left[0, 0, \frac{3}{4}\right]$	[1,6,18,21]
2	$\left[0, 0, \frac{1}{4}\right]$	[2,5,17,22]
3	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[3,8,20,23]
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[4,7,19,24]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,14,26,29]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[10,13,25,30]
7	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[11,16,28,31]
8	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[12,15,27,32]

Table 6: Wyckoff site: 8f, site symmetry: 2/m..

No.	position	mapping
1	$[0, 0, 0]$	[1,2,5,6]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,4,7,8]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[9,10,13,14]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[11,12,15,16]
5	$\left[0, 0, \frac{1}{2}\right]$	[17,18,21,22]
6	$\left[\frac{1}{2}, 0, 0\right]$	[19,20,23,24]
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,29,30]
8	$\left[0, \frac{1}{2}, 0\right]$	[27,28,31,32]

Table 7: Wyckoff site: 8g, site symmetry: mm'2'

No.	position	mapping
1	$\left[0, \frac{1}{4}, z\right]$	[1,6,28,31]
2	$\left[0, \frac{3}{4}, -z\right]$	[2,5,27,32]
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2} - z\right]$	[3,8,26,29]
4	$\left[\frac{1}{2}, \frac{3}{4}, z + \frac{1}{2}\right]$	[4,7,25,30]
5	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	[9,14,20,23]
6	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	[10,13,19,24]
7	$\left[0, \frac{3}{4}, \frac{1}{2} - z\right]$	[11,16,18,21]
8	$\left[0, \frac{1}{4}, z + \frac{1}{2}\right]$	[12,15,17,22]

Table 8: Wyckoff site: `16h`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	`[1,3]`
2	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}\right]$	`[5,7]`
4	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	`[6,8]`
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[9,11]`
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[10,12]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[13,15]`
8	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[14,16]`
9	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	`[17,19]`
10	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[18,20]`
11	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[21,23]`
12	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[22,24]`
13	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[25,27]`
14	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[26,28]`
15	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[29,31]`
16	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[30,32]`

Table 9: Wyckoff site: `16i`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ 0\right]$	`[1,19]`
2	$\left[\frac{3}{4},\ -y,\ 0\right]$	`[2,20]`
3	$\left[\frac{3}{4},\ y,\ \frac{1}{2}\right]$	`[3,17]`
4	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$	`[4,18]`
5	$\left[\frac{1}{4},\ -y,\ 0\right]$	`[5,23]`
6	$\left[\frac{1}{4},\ y,\ 0\right]$	`[6,24]`
7	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	`[7,21]`
8	$\left[\frac{1}{4},\ y,\ \frac{1}{2}\right]$	`[8,22]`
9	$\left[\frac{1}{4},\ y+\frac{1}{2},\ 0\right]$	`[9,27]`
10	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ 0\right]$	`[10,28]`
11	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[11,25]`
12	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[12,26]`
13	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ 0\right]$	`[13,31]`
14	$\left[\frac{3}{4},\ y+\frac{1}{2},\ 0\right]$	`[14,32]`
15	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[15,29]`
16	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[16,30]`

Table 10: Wyckoff site: `16j`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[1,18]`
2	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[2,17]`
3	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[3,20]`
4	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	`[4,19]`

continued ...

Table 10

No.	position	mapping
5	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[5,22]
6	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[6,21]
7	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[7,24]
8	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[8,23]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[9,26]
10	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[10,25]
11	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[11,28]
12	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[12,27]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,30]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[14,29]
15	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[15,32]
16	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[16,31]

Table 11: Wyckoff site: 16k, site symmetry: 2..

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2]
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[3,4]
3	$[-x,\ 0,\ 0]$	[5,6]
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[7,8]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[9,10]
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[11,12]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[13,14]
8	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[15,16]
9	$\left[x,\ 0,\ \frac{1}{2}\right]$	[17,18]
10	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[19,20]
11	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[21,22]
12	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[23,24]
13	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[25,26]
14	$\left[-x,\ \frac{1}{2},\ 0\right]$	[27,28]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[29,30]
16	$\left[x,\ \frac{1}{2},\ 0\right]$	[31,32]

Table 12: Wyckoff site: 16l, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{3}{4},\ 0,\ z\right]$	[1,20]
2	$\left[\frac{3}{4},\ 0,\ -z\right]$	[2,19]
3	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}-z\right]$	[3,18]
4	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	[4,17]
5	$\left[\frac{1}{4},\ 0,\ -z\right]$	[5,24]
6	$\left[\frac{1}{4},\ 0,\ z\right]$	[6,23]
7	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	[7,22]
8	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-z\right]$	[8,21]

continued ...

Table 12

No.	position	mapping
9	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	[9,28]
10	$\left[\frac{1}{4},\ \frac{1}{2},\ -z\right]$	[10,27]
11	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,26]
12	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[12,25]
13	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	[13,32]
14	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	[14,31]
15	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,30]
16	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[16,29]

Table 13: Wyckoff site: 16m, site symmetry: .m'.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	[1,31]
2	$\left[x,\ \frac{3}{4},\ -z\right]$	[2,32]
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[3,29]
4	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[4,30]
5	$\left[-x,\ \frac{3}{4},\ -z\right]$	[5,27]
6	$\left[-x,\ \frac{1}{4},\ z\right]$	[6,28]
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[7,25]
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[8,26]
9	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z\right]$	[9,23]
10	$\left[x+\frac{1}{2},\ \frac{1}{4},\ -z\right]$	[10,24]
11	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[11,21]
12	$\left[-x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[12,22]
13	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ -z\right]$	[13,19]
14	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	[14,20]
15	$\left[x,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[15,17]
16	$\left[x,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[16,18]

Table 14: Wyckoff site: 16n, site symmetry: m..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,6]
2	$[0,\ -y,\ -z]$	[2,5]
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[3,8]
4	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[4,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9,14]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10,13]
7	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,16]
8	$\left[0,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12,15]
9	$\left[0,\ y,\ z+\frac{1}{2}\right]$	[17,22]
10	$\left[0,\ -y,\ \frac{1}{2}-z\right]$	[18,21]
11	$\left[\frac{1}{2},\ y,\ -z\right]$	[19,24]
12	$\left[\frac{1}{2},\ -y,\ z\right]$	[20,23]

continued ...

Table 14

No.	position	mapping
13	$\left[\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[25,30]
14	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[26,29]
15	$\left[0, y+\frac{1}{2}, -z\right]$	[27,32]
16	$\left[0, \frac{1}{2}-y, z\right]$	[28,31]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x, -y, -z]$	[2]
3	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$[-x, y, z]$	[6]
7	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2}, y, \frac{1}{2}-z\right]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z\right]$	[9]
10	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[10]
11	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[11]
12	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-y, -z\right]$	[13]
14	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z\right]$	[14]
15	$\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[15]
16	$\left[x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[16]
17	$\left[x, y, z+\frac{1}{2}\right]$	[17]
18	$\left[x, -y, \frac{1}{2}-z\right]$	[18]
19	$\left[\frac{1}{2}-x, y, -z\right]$	[19]
20	$\left[\frac{1}{2}-x, -y, z\right]$	[20]
21	$\left[-x, -y, \frac{1}{2}-z\right]$	[21]
22	$\left[-x, y, z+\frac{1}{2}\right]$	[22]
23	$\left[x+\frac{1}{2}, -y, z\right]$	[23]
24	$\left[x+\frac{1}{2}, y, -z\right]$	[24]
25	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[26]
27	$\left[-x, y+\frac{1}{2}, -z\right]$	[27]
28	$\left[-x, \frac{1}{2}-y, z\right]$	[28]
29	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[30]
31	$\left[x, \frac{1}{2}-y, z\right]$	[31]
32	$\left[x, y+\frac{1}{2}, -z\right]$	[32]